SaaS Readiness App for Qlik Sense Enterprise Client-Managed

Configuration Guide

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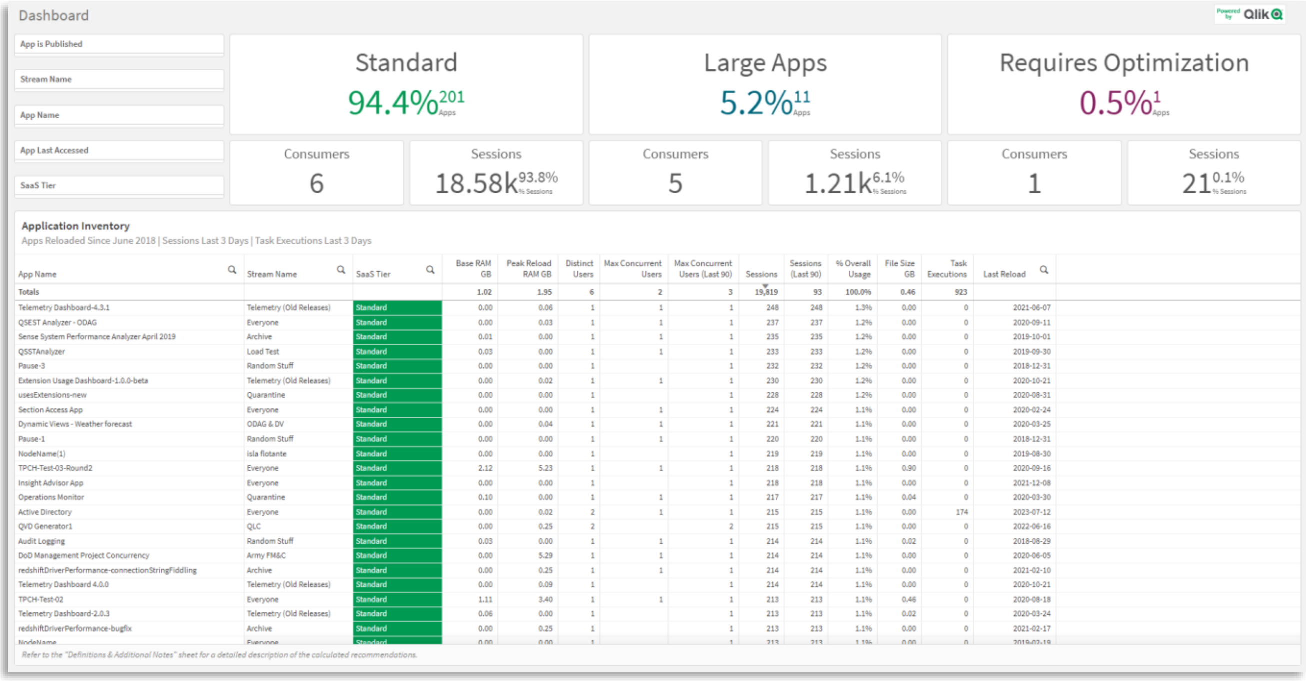
Disclaimer

These Tools provide general guidance regarding your organization's possible migration to Qlik Cloud. Estimates are based on information provided by Customer about your applications, data sources, data updates and expected usage patterns. The information provided by the Tools are general guidelines, and any assessment of migration paths to Qlik Cloud for your organization's needs will require additional due diligence.

The suggestions provided by the Tools are for informational purposes only, and do not represent a warranty or representation by Qlik of the actual Qlik Cloud deployment that may best suit your business requirements and preferences. The suggestions are based on the information provided by Customer, and any inaccuracy of, or changes to, the information provided by the Customer may result in significantly different evaluations and results. In the event of any conflict between the results of use of the ESR Tools and Qlik's standard Documentation, the terms of Qlik's standard Documentation shall control.

INTRODUCTION

This guide will cover the configuration of the SaaS Readiness application for Qlik Sense Enterprise Client-Managed. The SaaS Readiness application is a Qlik Sense application that is intended to be run on a Qlik Sense Enterprise Client-Managed site with the assistance of a Qlik Technical Field member or a Qlik Partner for existing clients that are considering Qlik Cloud. This application profiles the applications on a Qlik Sense Enterprise Client-Managed site and qualifies them into their appropriate Qlik Cloud tiers. The application also includes session usage, data connection metadata, as well as tasks and task cadences, all of which are important attributes of prioritizing and weighing the complexity of the client's migration of assets.



Pre-requisites

The high-level Qlik Sense Enterprise Client-Managed version requirements are as follows. Refer to numbered pre-requisites below for complete detail. At minimum, version June 2018 is required, however it is encouraged to leverage February 2020+ to gather lineage and reload data, as these are key data points in discerning the ease in which an application can be fully migrated to Qlik Cloud.

	June 2018+ (mandatory)	June 2019+	February 2020+ (suggested)
App Metadata	✓	✓	✓
App Lineage		✓	✓
Peak Reload RAM			✓

Mandatory

1. Qlik Sense Enterprise Client-Managed June 2018+

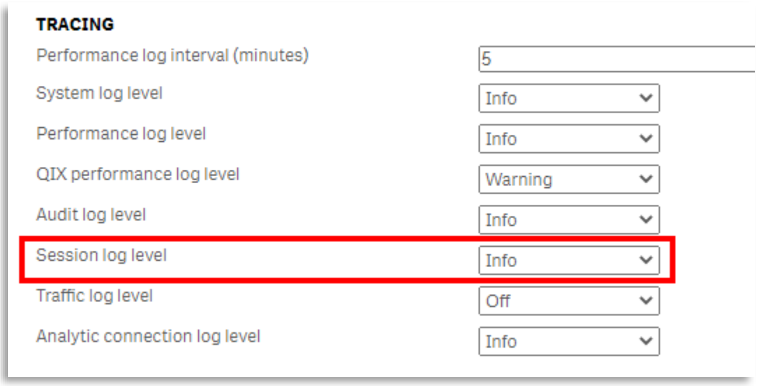
   The core requirement of this application is to profile applications into their respective Qlik Cloud tiers, which is made possible by the application metadata endpoint, introduced in June 2018:

   `http(s)://{FQDN}/api/v1/apps/{APP_GUID}/data/metadata`

   This endpoint provides the base RAM footprint (the application's resting RAM footprint without any session/result set caching) that is written out post-application reload.

2. Session Log Level (Info)

   The **Session log level** must be set to **Info** on all Engines. This setting is default on all Qlik Sense Enterprise Client-Managed installations. Session usage information per application is integral to weighing and prioritizing the migration of applications to Qlik Cloud, as applications that are highly used are typically of greater priority, and commonly have larger impacts on hardware savings.

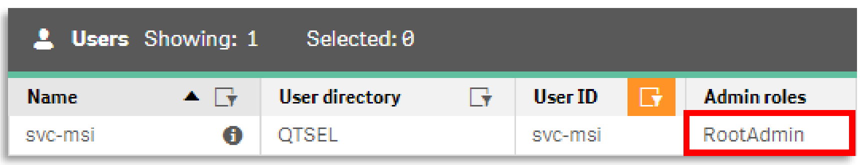


3. Task Execution Log Level (Info)

   The **Task execution log level** must be set to **Info** on all Schedulers. This setting is default on all Qlik Sense Enterprise Client-Managed installations. Task execution information is helpful for clients that are interested in migrating applications to dedicated capacity, as it provides the application the ability to calculate maximum task concurrency along with maximum concurrent peak reload RAM.

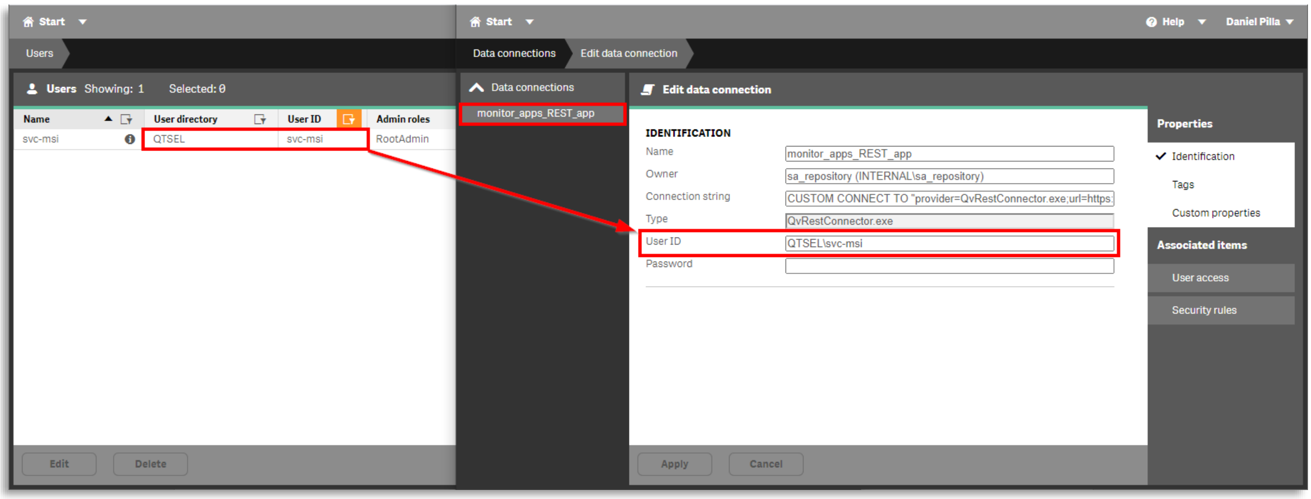
4. Qlik Service Account has Role of RootAdmin

   The Qlik service account must have the **RootAdmin** role, as this application gathers metadata across licenses, apps, app metadata, app lineage, tasks, nodes, and data connections.



5. Validate Qlik Service Account is the User Account Associated with Monitoring Connections

The Qlik service account needs to be the same account found in the **User ID** field of the `monitor_apps_REST_app` data connection. If this has been modified to another user, ensure that the user is also a **RootAdmin** as per the step above.



6. A Virtual Proxy with Windows Authentication Enabled

A **Virtual Proxy** with Windows authentication must be enabled, as this application leverages the **monitor_apps_REST_app** data connection that ships with the product, which uses Windows authentication over the Qlik service account. Forms must not be enabled.

If a virtual proxy with Windows authentication does not exist, an additional can be created if necessary. This is a requirement for all of Qlik's monitoring applications as well (Operations Monitor, License Monitor, etc). If the monitoring apps are not currently operational, refer to this Qlik Support Knowledge Base Article. In addition, do note that Kerberos is not supported.

Non-Critical but Highly Suggested

1. Qlik Sense Enterprise Client-Managed June 2019+

   In Qlik Sense Enterprise Client-Managed June 2019, a new RESTful endpoint was added that exposes lineage data per-application.

   `http(s)://{FQDN}/api/v1/apps/{APP_GUID}/data/lineage`

   This application considers the data connection types found from this endpoint and weighs them as part of a calculation in the **SaaSMigration** field. For instance, if an application only uses the Salesforce connector and QVDs, it is likely easily portable to Qlik Cloud (spare other checks that are performed like reload cadence, task chains, etc).

2. Qlik Sense Enterprise Client-Managed February 2020+

   In Qlik Sense Enterprise Client-Managed February 2020, an additional attribute was included in the application metadata endpoint noted above which documents the **Peak Reload RAM**. This is the maximum amount of RAM the application took while reloading, which is important to note as there are in fact throttled thresholds in Qlik Cloud on reload capacity (varies by tier). This data is only relevant for applications that are being physically migrated to Qlik Cloud, as opposed to just being distributed there for consumption.

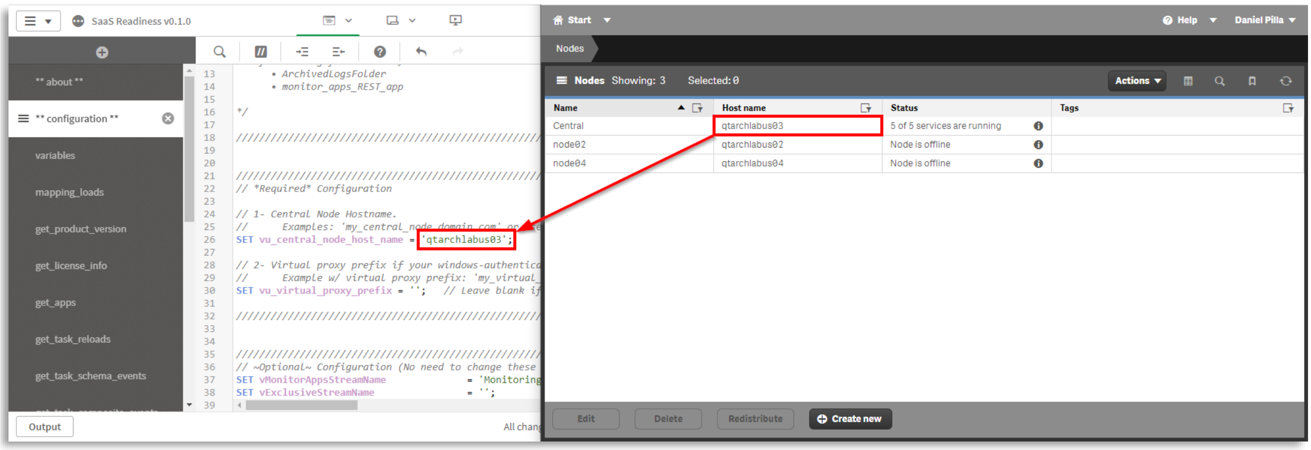
Setup

Configure Script Variables

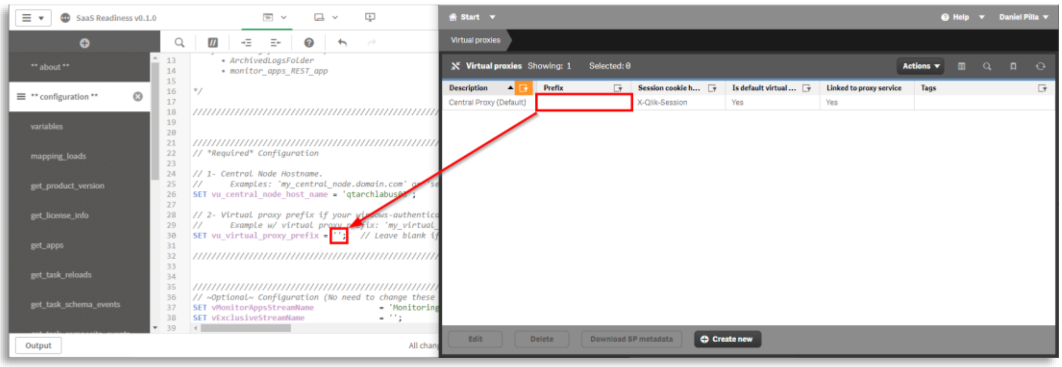
1. Import the SaaS Readiness app into the customer's Qlik Sense Enterprise Client-Managed site.

2. Open the application and navigate to the ****configuration**** tab in **Data Load Editor**.

3. Find the `vu_central_node_host_name` variable and set it to either the **FQDN** or **machine name** of the site's **Central Node**. If this is a single node site, the value of `localhost` may also be used. As this application needs to gather metadata from *all* applications, using the central node ensures this is possible.



4. Find the `vu_virtual_proxy_prefix` variable. If the prefix-less **Virtual Proxy** linked to the central node is using **Windows Authentication** (without forms enabled), leave this blank (this is the default). If there is a prefix set for the virtual proxy using Windows authentication, enter it here (e.g. `SET vu_virtual_proxy_prefix = 'prefix';`. Be sure to not include any `/` characters in the string.



5. **(Optional)** `vMonitorAppsStreamName` – this is the default name of the monitoring apps stream. If it has been changed, modify it here. This is so all monitoring apps are excluded from analysis, as they have no relevance in Qlik Cloud.

6. **(Optional)** `vSessionLogDaysBack` – this variable is further documented below, however note that it controls the amount of session logs that are brought in (e.g. only the last 90 days). If the Qlik Sense Enterprise Client-Managed site is many years old, or if the goal is to check if the application successfully reloads (trial reloading), set this value to something lower, e.g. `30`. This will be set in the **Execution** section below.

7. **(Optional)** `vTaskLogDaysBack` – this variable is further documented below, however note that it controls the amount of task execution logs that are brought in (e.g. only the last 90 days). If the Qlik Sense Enterprise Client-Managed site is many years old, or if the goal is to

check if the application successfully reloads (trial reloading), set this value to something lower, e.g. `30`. This will be set in the **Execution** section below.

8. **(Optional)** `vAppLastReloadedDaysBack` – this variable is further documented below, however note that it controls the amount of applications that are scanned for metadata (e.g. only applications that have been reloaded in the last 90 days). As application metadata only became available as of the June 2018 release of Qlik Sense Enterprise Client-Managed, the default date minimum date for an application to be reloaded is June 15th, 2018 (roughly the release date). This variable allows that time to be overridden to something more recent. This is helpful for either trial reloading the application or for reloading the application in batch (e.g. reloading with it set to 30, then 90, then 180, and so on). This variable will be explored more in the **Execution** section below.

Execution

Trial Reload (Subset of Data)

First and foremost, the application should be reloaded with a limited set of data to confirm successful operation, as this application take multiple hours to reload on large sites.

Adjust Variables

Keep the following variables at the same value. For instance, if `7` is chosen, use that for the following variable values.

1. Find the `vSessionLogDaysBack` variable, and set it to a small number of days as to only fetch data from a minimal amount of session logs, as this will allow the application to reload more quickly, e.g.

   ```
   SET vSessionLogDaysBack = 7;
   ```

2. Find the `vTaskLogDaysBack` variable, and set it to a small number of days as to only fetch data from a minimal amount of task execution logs, as this will allow the application to reload more quickly, e.g.

   ```
   SET vTaskLogDaysBack = 7;
   ```

3. Find the `vAppLastReloadedDaysBack` variable, and set it to a small number of days as to only fetch application from applications that have been reloaded within the last x days (the current time – the variable), as this will allow the application to reload more quickly, e.g.

   ```
   SET vAppLastReloadedDaysBack = 7;
   ```

Trial Reload Considerations

There are two workflows in which the application can be reloaded:

A. Reload as a task.

B. Reload from the Hub.

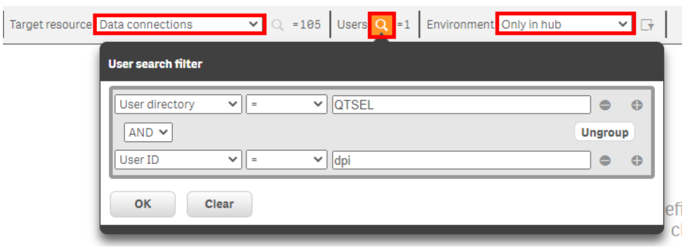
Reloading from the Hub (B) while seemingly easier and more straightforward requires the likely modification of security rules, thereby the suggested approach is to always reload as a task (A). If reloading from the Hub is desired, that setup is documented below.

Trial Reload as a Task (Option A)

1. Navigate to the **Tasks** section of the **QMC**.
2. Select **Create new**.
3. Enter a **Name** and select the SaaS Readiness application.
4. Select **Apply** as there is no need to set a schedule for this application.
5. Lastly, select the newly created task, and select **Start**.

Trial Reload from the Hub (Option B)

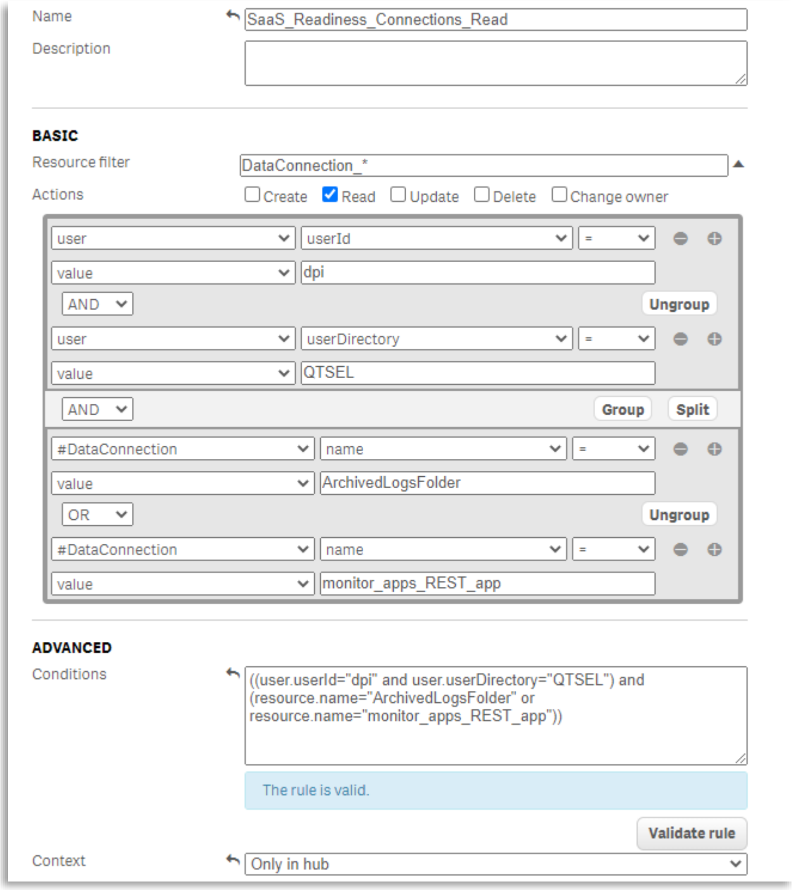
1. Navigate to the **Audit** section of the **QMC**.
2. Select **Data connections** as the **Target resource**.
3. Find the user that will be reloading the application from the Hub and enter the User ID or other attributes by which you'd like to identify the user by.
4. Under **Environment** select **Only in Hub**.



5. Select **OK** and then **Audit**.
6. Select **Transpose**.
7. The user will need **Read** access to both `ArchivedLogsFolder` (for the session logs) as well as `monitor_apps_REST_app` (for all of the QRS API calls). Ensure that the user has

this access. If they do not, please refer back to Option A, or consult with the site's Qlik administrator(s). An example security rule to grant access could resemble the following:

```
((user.userId="dpi" and user.userDirectory="QTSEL") and
(resource.name="ArchivedLogsFolder" or
resource.name="monitor_apps_REST_app"))
```



8. After confirming and/or granting access, ensure that the user can see both data connections in the Hub. This occasionally can take a restart of the repository service.

9. Reload the application from the Hub.

Full Reload Options

After confirming the trial reload is successful, there are two options in which the application can be reloaded. Given the number of applications in the site, choose one or the other:

- **In batches**
    - This option is best for Qlik Sense Enterprise Client-Managed sites that have greater than 500 applications. It allows the application to be reloaded in progressively larger chunks while taking advantage of incremental loads using QVDs.
- **In a single reload (full)**
    - This option is best for Qlik Sense Enterprise Client-Managed sites that have a maximum of 500 applications. The application will fetch metadata for all applications at once.

Batch Reload

1. Navigate to the Hub and open the SaaS Readiness application.
2. After having reloaded the application in a trial reload as per the Trial Reload (Subset of Data) section of this document, confirm that the `vSessionLogDaysBack` and `vTaskLogDaysBack` variables are both set to a low number, e.g. `7`. These logs will be loaded in bulk on the very last reload.

    ```
    SET vSessionLogDaysBack = 7;
    SET vTaskLogDaysBack = 7;
    ```

3. Find the `vAppLastReloadedDaysBack` variable. If it was set to `7` on the trial run, set it to a higher number, say `30`.

    ```
    vAppLastReloadedDaysBack = 30;
    ```

4. Reload the application either as a task or from the Hub, depending on how long the trial reload took.
5. Repeatedly follow steps 3 and 4 by continuing to increment the `vAppLastReloadedDaysBack` variable. If it was set to `30` on the last run, set it to a higher number, say `90` (these chunks can get larger on each iteration, as there *should* be less applications that haven't been reloaded the further back the variable is set). Continue this process up until the desired number of days has been reached, say `365`, or after reaching `365`, set the variable to `-1` to reload all of the remaining metadata for all applications (back to June 15th, 2018).
6. Lastly, set the `vSessionLogDaysBack` and `vTaskLogDaysBack` variables to the desired range, e.g. `180`.

    ```
    SET vSessionLogDaysBack = 180;
    SET vTaskLogDaysBack = 180;
    ```

7. Reload the application for the last time.

Full Reload

1. Navigate to the Hub and open the SaaS Readiness application.

2. After having reloaded the application in a trial reload as per the Trial Reload (Subset of Data) section of this document, confirm that the `vSessionLogDaysBack` and `vTaskLogDaysBack` variables are both set to the desired number, e.g. `180`.

   ```
   SET vSessionLogDaysBack = 180;
   SET vTaskLogDaysBack = 180;
   ```

3. Find the `vAppLastReloadedDaysBack` variable. Set it to `-1` (which turns off the custom range) or `365` if only a year of data is desired.

   ```
   vAppLastReloadedDaysBack = -1;
   ```

4. Follow Option A from **Trial Reload** above to reload the application from a **Task**, as the full reload can take multiple hours on larger sites.

Potential 404 Errors

If the application is throwing 404 errors (not found) for either the metadata or lineage endpoints, after confirming a successful Trial Reload (Subset of Data), set the `ErrorMode` variable to `0` to move through them. These errors can be caused by several difference scenarios (e.g. an app being deleted mid-reload), but should not be common and can be moved through without any significant impact to the application.

```
SET ErrorMode = 0;
```

LEAD WITH DATA

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